Toward Cross-Satellite Change Detection: A Spatial–Spectral Dual-Alignment Joint Learning Network

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***Abstract*—Despite boosting change detection (CD) performance, most deep neural networks (DNNs) are designed for multitemporal remote sensing images acquired by the same satellite. However, it is difficult for a single satellite to acquire continuous data for high-frequency CD over a large area. Although it is feasible to conduct CD on multitemporal remote sensing images from different satellites, differences in spectral response and spatial resolution cannot be ignored. To mitigate these issues, we propose a spatial–spectral dual-alignment joint learning network (SSDA-Net). The proposed framework introduces a dual-resolution joint learning strategy that exploits LR features from the super-resolution (SR) network to guide HR feature alignment in the CD network, thereby suppressing spectral inconsistencies caused by different sensor responses and spatial mismatches induced by resolution differences between satellites. Specifically, for multitemporal inputs with different resolutions, we first send the LR one into the SR network to improve its spatial resolution, and then the super-resolved result and the HR one are fed into the CD network to extract HR features. To exploit the spatial and spectral information in both LR and HR features, we design dual-alignment learning modules (DALMs), which are composed of a spectral alignment block (SpeAB) and a spatial alignment block (SpaAB). Through the two modules, the spectral and spatial differences between multitemporal images from different satellites are suppressed. Finally, features from DALMs are used to generate the change map. To validate the effectiveness of the proposed SSDA-Net, we construct a real cross-satellite change detection (SD-CSCD) dataset. Experimental results on the newly built SD-CSCD dataset, the DE-CD dataset, and a synthesized dataset demonstrate that our SSDA-Net achieves better performance than state-of-the-art methods. The SD-CSCD dataset and the code of SSDA-Net are available at https://github.com/XxxJ0909/SSDA-Net**

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I. Introduction

CHANGE detection (CD) aims to automatically identify regions of change by comparing multitemporal remote sensing images [1], [2]. This technology plays a pivotal role in many applications, such as land cover monitoring [3], urban sprawl analysis [4], and forest resource assessment [5]. With the increasing number of Earth observation satellites, high-resolution (HR) remote sensing images are acquired almost continuously, which further boosts the performance of CD [6].

Over the past few decades, CD paradigms have evolved from traditional methods to deep neural network (DNN)-based methods [7], [8]. Within traditional paradigms, change vector analysis (CVA) [9], multivariate alteration detection (MAD) [10], and their variants have been extensively studied. Meanwhile, sparse representation models have also been applied to CD tasks [11]. However, the performance of these methods is restricted by their limited representation capability. Due to their powerful nonlinear modeling capacity, DNN-based methods have attracted extensive attention. For example, convolutional neural networks (CNNs) [12], generative adversarial networks (GANs) [13], transformers [14], [15], and diffusion models [16] have been increasingly adopted to detect change regions in multitemporal images and have achieved considerable performance improvements. Nevertheless, the majority of these methods focus only on change analysis for multitemporal images acquired from the same satellite, as shown in Fig. 1(a).

With the increasing demand for high-frequency or short-interval change analysis over large areas, it is difficult to obtain the required data when relying solely on a single satellite. Thus, researchers have begun to shift their attention to cross-satellite change detection (CSCD), in which multitemporal images are collected by different imaging sensors and exhibit distinct spectral and spatial properties. However, since CSCD images are difficult to acquire, register, and label, some studies resort to simulating multitemporal data with different spatial resolutions by downsampling single-temporal images from benchmark datasets, such as LEVIR-CD [17], as shown in Fig. 1(b). Thus, the image pairs generated in this manner are referred to as cross-resolution change detection (CRCD) data. For CRCD, Zheng et al. [18] employed a cross-resolution difference learning strategy to generate pixelwise difference maps for inferring change information. Liu et al. [19] combined

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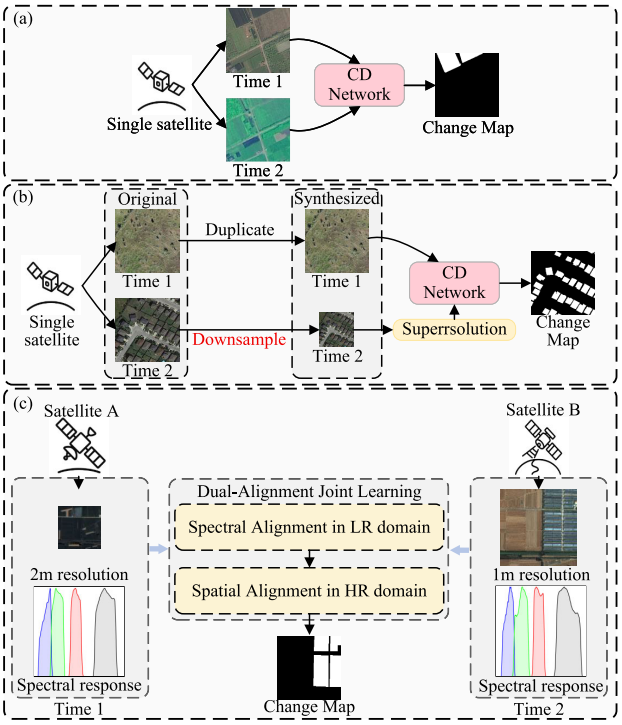


Fig. 1. Schematic of different CD tasks. (a) CD on multitemporal images acquired by the same satellite. (b) CD on cross-resolution multitemporal images. (c) CD on multitemporal images acquired by the different satellites.

a GAN-based super-resolution (SR) module with a stacked attention mechanism to align the spatial resolutions of multitemporal images and produce change maps. Zhang et al. [20] proposed a semantic-aware network to alleviate the adverse effects of resolution mismatches on CD results. Although good results have been achieved, the synthetic data generation process ignores the intrinsic differences in spectral and spatial response between satellites in cross-satellite scenarios.

Recently, Chen et al. [21] constructed a CSCD dataset, named DE-CD, based on the original data for the daily semantic change segmentation task [22] and utilized an implicit neural representation network to handle the resolution differences in the data. Subsequently, Li et al. [23] utilized a dual-branch network to integrate the LR and HR change information for improved change estimation. However, the spatial resolution of multitemporal images in the DE-CD dataset is limited. In this dataset, the preevent images were acquired from the Sentinel-2 satellite, with a spatial resolution of 10 m, while the postevent images were provided by Planet, with a spatial resolution of 3 m. Therefore, this dataset is not suitable for supporting CD on very-high-resolution (VHR) data, such as imagery with a spatial resolution of 1 m. Moreover, similar to methods designed for synthetic multitemporal data, the above two networks still primarily address the issue caused by spatial resolution differences, without considering the impact of spectral differences. As shown in Fig. 1(c), different satellites have distinct imaging sensors with different spatial resolutions and spectral response characteristics; consequently, the multitemporal images they acquire also differ in these two aspects. For example, images taken by different sensors over the same area contain not only spectral differences caused by time and season but also variations caused by different spectral responses, which, if not mitigated, may lead to pseudochanges in CD results.

It is worth noting that some Google Earth-based CD datasets may also contain imagery from different sensors. However, they are typically provided as coregistered image pairs with the same spatial resolution, so most methods still treat them as homogeneous inputs. By contrast, real CSCD explicitly involves both spatial-resolution mismatch and sensor-dependent spectral inconsistency, which requires dedicated modeling. To deal with the aforementioned issues, we first construct a VHR CSCD dataset, called SD-CSCD, which covers Shandong Province in China. In this dataset, preevent images with a spatial resolution of 2 m were acquired by the GF-6 satellite in 2019, and postevent images with a spatial resolution of 1 m were acquired by the GF-2 satellite in 2024. We then propose a spatial–spectral dual-alignment joint learning network (SSDA-Net) to effectively model and alleviate spectral and spatial discrepancies between satellites for more accurate change analysis. Specifically, the proposed method is composed of an SR network and a CD network. For each low-resolution (LR) and HR image pair in multitemporal data, the LR image is fed into the SR network to generate its super-resolved counterpart. For the CD network, the super-resolved image and the HR image are used as inputs to extract change features. To efficiently suppress the differences in terms of spectral and spatial properties, we design the dual-alignment learning modules (DALMs) consisting of a spectral alignment block (SpeAB) and a spatial alignment block (SpaAB). The SpeAB employs the spectral difference attention derived from a Gram matrix to harmonize the spectral differences in LR features from the SR network. In SpaAB, the aligned LR features from the SpeAB are used to guide the integration of HR features from the CD network via a bilateral coupling mechanism. Meanwhile, to produce satisfactory super-resolved images, we adopt a joint supervised–unsupervised learning strategy to train the SR network, which also helps mitigate discrepancies in data properties across satellites. Finally, we conduct extensive experiments on our constructed SD-CSCD dataset, the DE-CD dataset, and the synthesized LEVIR-CD dataset. The results demonstrate the effectiveness of our proposed SSDA-Net compared with nine state-of-the-art methods.

The main contributions of this article are summarized as follows.

1) We propose an SSDA-Net to suppress the influences resulting from the spatial and spectral differences in cross-satellite data, in which LR features from SR networks and HR features from the CD network are integrated by the dual-resolution joint learning strategy.
2) We design a bilateral coupling mechanism in SpaAB. Through the mechanism, the HR features from different times are aligned better with the guidance of LR features.
3) We create a novel VHR CSCD dataset, named SD-CSCD, consisting of 7280 bitemporal image pairs. Among them, the 2-m-resolution images are acquired by the GF-6 satellite; they are the preevent images and have a size of 128 $\times$ 128 pixels. The postevent images

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are collected by the 1-m-resolution GF-2 satellite, and have a size of 256 × 256 pixels.

4) Extensive experiments on three datasets demonstrate that our method achieves state-of-the-art performance, validating its superior accuracy and effectiveness in challenging cross-satellite scenarios.

The remainder of this article is organized as follows. Section II provides a comprehensive review of related work. Section III presents the proposed SSDA-Net in detail. Section IV introduces the experimental settings and provides extensive comparison results on different datasets. Finally, Section V concludes the article.

II. Related Work

A. DNN-Based CD

At present, the development of DNNs has dramatically improved CD performance, and various network architectures have been applied to this field. For example, Fang et al. [24] incorporated dense skip connections into the channel attention network to enhance the detection of small-scale changes. Liu et al. [25] devised a difference attention module to capture the style shifts in the spectral domain. Liu et al. [26] introduced a supervised adaptive framework, leveraging a GAN to mitigate the impact of pseudochanges on CD accuracy. Hou et al. [27] proposed a GAN-based method that effectively modeled change generation to mitigate feature inconsistency in bitemporal images. All these methods, although they obtained good results, suffer from some issues in terms of learning of global spatial dependencies in large scenes.

Recently, transformers have attracted widespread attention from researchers due to their long-range modeling ability. For instance, Pan et al. [28] proposed a Swin-Transformer-based CD method, which enhanced the modeling of contextual features. Yang et al. [29] combined a CNN and a transformer to simultaneously learn the local and global information in multitemporal images. Furthermore, Zhang et al. [30] considered the relation changes in multitemporal images and built a cross-temporal difference attention for CD. Liu et al. [31] integrated multiscale features for better offset calibration, which avoided the pseudochange caused by side-looking issues. Similar to Transformer, Mamba [32] has also been used for change analysis because of its global learning capacity. Liu et al. [33] proposed a Mamba-based spatiotemporal interaction method, which merged the global information from multiple dimensions for better performance. Chen et al. [34] proposed a state-space-model-based method for remote sensing CD, which enhances global contextual modeling and spatiotemporal feature interaction to alleviate the limitations of existing methods. Zhou et al. [35] also employed Mamba and introduced the saliency mechanism to balance the changed and unchanged features.

Recently, diffusion models have emerged as a powerful approach capable of modeling complex data distributions, offering a new paradigm for CD. For example, Wang et al. [36] utilized the diffusion model to generate pseudochange pairs for data augmentation. Li et al. [37] proposed a diffusion-guided CD method, in which spatial and frequency information were jointly exploited for more reliable feature extraction. Zhu et al. [38] designed a diffusion-based wavelet attention network to extract the multilevel differential features in images more efficiently. Han et al. [39] introduced the hierarchical representation into the diffusion model to alleviate multiscale change ambiguity in images. Zhu et al. [40] embedded a graph convolution network into the diffusion model to better capture the topological relationships in images. Currently, an increasing number of researchers are turning their attention to applying visual fundamental models to CD tasks. For example, Ding et al. [41] proposed a segment anything model-based method for very HR remote sensing CD, which adapted pretrained visual representations to remote sensing scenes through task-oriented change aggregation and semantic representation learning.

B. DNN-Based CSCD

Although the aforementioned CD methods have demonstrated notable improvements in accuracy with respect to previous generation approaches, they have been primarily developed for multitemporal images acquired by the same satellite. In practical high-frequency or short-interval monitoring scenarios, multitemporal images are often acquired by different satellite sensors, leading to discrepancies in terms of spatial resolution and spectral characteristics. Consequently, recent research has increasingly shifted toward CRCD and CSCD. In CRCD, the data are obtained by downsampling the images from only one temporal phase in existing benchmark datasets to create multitemporal image pairs with different spatial resolutions. For example, Li et al. [42] proposed an iterative SR CD method using coarse–fine resolution images to produce fine-scale land cover changes. Tian et al. [43] used a lightweight SR network and built a CD network consisting of deformable, dilated, and attention modules for better cross-resolution alignment. Similarly, Li et al. [44] adopted an SR network to improve the resolution of LR images and extracted change information by contrastive learning. Very recently, Ng et al. [45] applied the segment anything model fine-tuned by automatic prompts from multiscale features to CRCD. In addition, Ning et al. [46] leveraged Fourier and wavelet transforms to produce resolution-invariant features, thereby mitigating resolution-induced distortions. Wang et al. [47] incorporated the continuous-resolution representation and resolution-difference modulation to improve the estimation of changed areas. Meanwhile, Liu et al. [48] constructed a CD model for multitemporal images with arbitrary resolution differences, in which a magnification-arbitrary network was used to match the resolutions between images. In parallel, heterogeneous CD (HeCD) methods have also attracted attention as a label-free alternative for handling multitemporal images from heterogeneous sensors. Sun et al. [49] designed a HeCD method for unregistered image pairs that mitigates misregistration through global mapping and local searching. Sun et al. [50] proposed a network that inferred changes from intraimage relationships without explicit cross-image comparison. Although the CRCD task has been explored, there is still a gap in practical application owing to the synthetic cross-resolution data. Currently, a few studies have begun to focus on CSCD in

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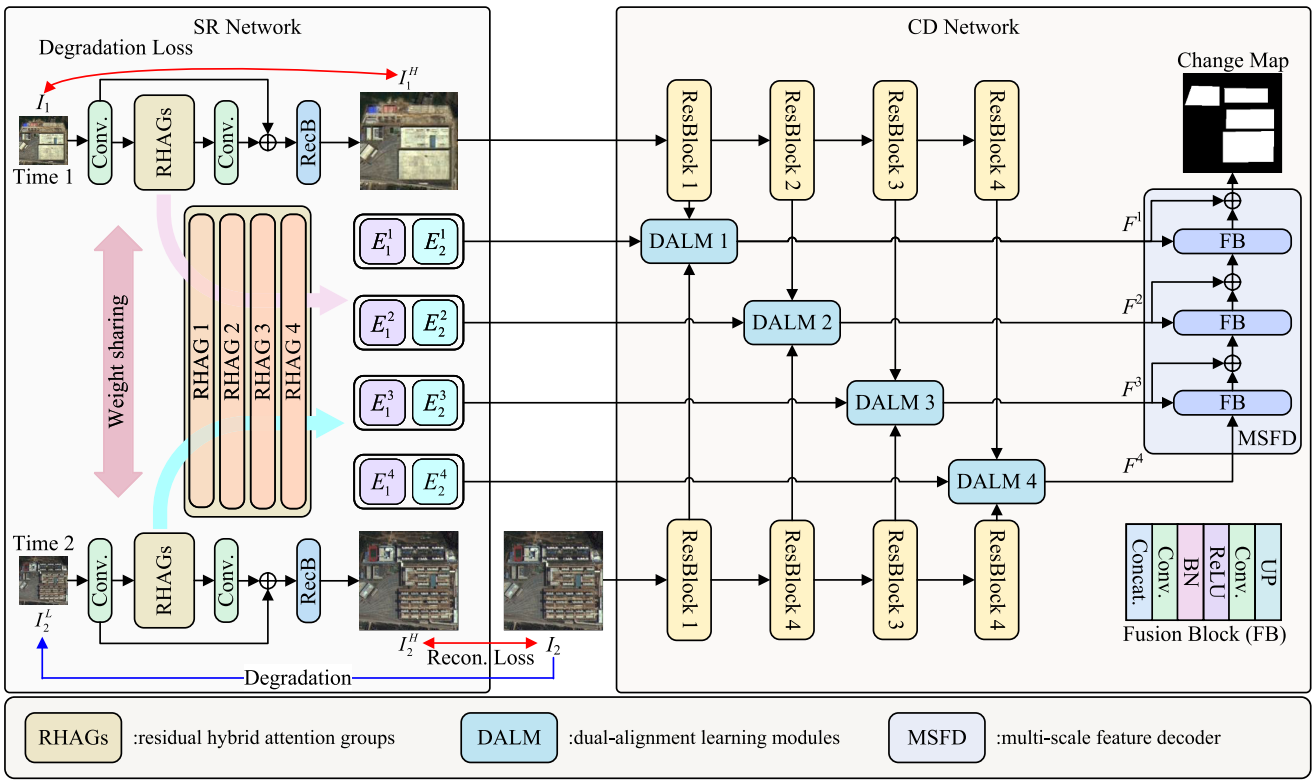


Fig. 2. Architecture of the proposed SSDA-Net.

real scenarios. For instance, Chen et al. [21] and Li et al. [23] conducted CD comparisons on the DE-CD dataset. In addition, Yang et al. [51] introduced a hierarchical window aggregation network that directly aligned the representations of cross-satellite data in latent space. However, the performance of these methods on the DE-CD benchmark dataset may still need improvement, possibly because this dataset was constructed using datasets from other tasks. Meanwhile, these methods considered SR networks and CD networks separately, without jointly utilizing the LR features in SR networks and the HR ones in CD networks. This prevents the effective alignment of spatial and spectral information from cross-satellite data, affecting CD performance. To achieve joint learning of SR and CD networks, we propose SSDA-Net to effectively exploit the LR and HR features to reduce pseudochanges resulting from spectral and spatial differences.

III. Proposed Method

In this section, we first present the overall architecture of SSDA-Net. Subsequently, we provide detailed descriptions of its two main components, i.e., the SR network and the CD network.

A. Overall Architecture

Fig. 2 illustrates the overall architecture of the proposed SSDA-Net. The bitemporal inputs of SSDA-Net are a cross-satellite image pair $\{I_1 \in \mathbb{R}^{h\times w\times c}, I_2 \in \mathbb{R}^{rh\times rw\times c}\}$. h and w denote the spatial dimensions of the LR image I_1 from Time 1. The size of the HR image I_2 from Time 2 is $rh \times rw$, and r is the resolution ratio between I_1 and I_2. c is the number of bands in images. First, we simplify the hybrid attention transformer (HAT) [52] and use it as the SR network to improve the resolution of I_1. Due to the lack of the HR version of I_1 as ground truth for training, we downsample I_2 with a ratio of r to obtain its LR counterpart $I_2^L \in \mathbb{R}^{h\times w\times c}$, and then the SR network is trained on the paired images. Meanwhile, to ensure the transferability of the SR network on data from Time 1, the degradation relationship between I_1 and I_1^H is also considered so that the SR network is jointly trained on multitemporal images. Then, for the extraction of the change information, the super-resolved result $I_1^H \in \mathbb{R}^{rh\times rw\times c}$ of I_1 and I_2 from Time 2 are sent into the CD network whose backbone is a modified ResNet-18. In order to sufficiently exploit the LR features in the SR network, we also introduce them into the learning of change information. Considering the spectral and spatial differences between cross-satellite data, we design DALMs to better align the changed and unchanged information between multitemporal images in feature space. The DALM is composed of a SpeAB and a SpaAB, which employ the spectral difference attention and the bilateral coupling mechanism for difference suppression, respectively. Finally, all features from DALMs are integrated into the multiscale feature decoder (MSFD) for the generation of the change map.

B. SR Network

In the proposed SSDA-Net, an SR network is needed to improve the resolution of the LR image at Time 1. Here, we

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utilize the distinguished HAT as the SR network. Considering the tradeoff between SR performance and computational complexity, we reduce the number of residual hybrid attention groups (RHAGs) from 6 to 4, and meanwhile cut the number of hybrid attention blocks in RHAG to half of the original. Finally, the desired HR image is generated by a reconstruction block (RecB) composed of convolution layers and a pixel shuffle operation.

For the cross-satellite data, I_1 has no corresponding HR ground truth for training the SR network, while I_2 is an HR image. Hence, we downsample I_2 to the size of I_1 and employ the image pair $\{I_2, I_2^L\}$ to train the SR network. However, the SR network, only trained on the data from Time 2, may suffer from performance degradation when directly applied to the LR image at Time 1 due to the spectral and spatial differences between different times. So, we further incorporate the degradation relationship among I_1 and I_1^H for the joint learning of the SR network.

According to the above formulation, we employ the L1 loss to measure the reconstruction error of I_2^H, and the degradation loss between I_1 and I_1^H is also computed. The L1 loss is adopted as the reconstruction objective following the original HAT design, as it better preserves high-frequency spatial details. The loss is defined as

$$\mathcal{L}_{\mathrm{SR}} = \lambda \left\| I_1 - D\left(I_1^H\right) \right\|_1 + \left\| I_2 - I_2^H \right\|_1 \tag{1}$$

where I_2^H is the SR result of I_2^L. D and λ denote the degradation operator and the tradeoff parameter, respectively. Through the joint training of the SR network, the performance reduction caused by the property shift is avoided. Finally, the LR images from different temporal phases share the same SR model for resolution improvement.

C. DALMs

To properly use the spectral and spatial information in cross-satellite data, LR features from the SR network and HR features from the CD network are simultaneously considered for the prediction of the change map. So, we design the DALM to achieve joint learning on these features. In DALM, SpeAB is imposed on the features corresponding to LR bitemporal images to deal with the effects of spectral discrepancies. Then, the aligned spectral feature from SpeAB is further utilized to guide the integration of features of HR images in the CD network. Through the joint learning of LR and HR features, the differences in terms of spectral response and spatial resolutions are effectively suppressed.

1) Spectral Alignment Block (SpeAB): For multitemporal images from different satellites, the differences in spectral information stem not only from different seasons and illumination conditions, but also from distinct spectral responses. These factors result in multitemporal images having inconsistencies in terms of spectral style, which are detrimental to CD. Inspired by the modeling of style information [53], [54], we construct the SpeAB to align the spectral differences in images. In contrast to style transfer, we improve the spectral consistency by aligning the Gram matrix.

Specifically, as illustrated in Fig. 3(a), the bitemporal LR features $E_1^i \in \mathbb{R}^{(r^2hw)\times c}$ and $E_2^i \in \mathbb{R}^{(r^2hw)\times c}$ from the ith RHAG block are first projected by a convolution block and reshaped as $R_1^i \in \mathbb{R}^{rh\times rw\times c}$ and $R_2^i \in \mathbb{R}^{rh\times rw\times c}$ for the calculation of Gram matrix

$$\begin{aligned} G_1^i &= \left(R_1^i\right)^T R_1^i \\ G_2^i &= \left(R_2^i\right)^T R_2^i \end{aligned} \tag{2}$$

where the size of G_1^i and G_2^i is $c \times c$, and c is the number of channels. The superscript i indicates the index of the RHAG block and is also the index of scale in DALM. Then, the spectral difference attention derived from the difference between G_1^i and G_2^i is used to mitigate the spectral mismatches in multitemporal features, which is formulated as

$$\begin{aligned} P_1^i &= R_1^i \left(I + G_1^i - G_2^i\right) \\ P_2^i &= R_2^i \left(I + G_2^i - G_1^i\right) \end{aligned} \tag{3}$$

where I is an identity matrix with a matched size. Finally, the aligned spectral features P_1^i and P_2^i are reshaped and concatenated to obtain the output S^i of the ith SpeAB. Through the spectral compensation, SpeAB can prevent the proposed SSDA-Net from producing pseudochange results due to different imaging conditions and spectral response.

2) Spatial Alignment Block (SpaAB): In DALM, SpaAB aims to further harmonize the HR features F_1^i and F_2^i from different times and integrate them for the generation of the change map. As shown in Fig. 3(b), we first use the aligned LR feature as the guidance and implement the cross-resolution attention (CRA) on the inputs of SpaAB

$$\begin{aligned} X &= \mathrm{Softmax}\left(\frac{Q_1 K_L^T}{\sqrt{d}}\right) V_L \\ Y &= \mathrm{Softmax}\left(\frac{Q_2 K_L^T}{\sqrt{d}}\right) V_L \end{aligned} \tag{4}$$

where K_L and V_L are the key and value of the LR feature S^i, respectively. Q_1 and Q_2 correspond to F_1^i and F_2^i, respectively. d stands for the feature dimension. Considering the gradual reduction in terms of the size of F_1^i and F_2^i, we directly view each pixel in them as a token, while S^i at the ith scale is partitioned into patches with a size of $2^{i-1} \times 2^{i-1}$ for its embedding. Then, X and Y are combined with the results of multihead self-attention (MSA) on F_1^i and F_2^i or feature enhancement, respectively.

To further align the HR features from different times, we design a bilateral coupling mechanism by estimating the multitemporal HR features O_1 and O_2, which are obtained after CRA and MSA. Specifically, O_1 and O_2 are first calculated and normalized along both column and row to produce bilateral similarity matrices. Then, they are refined by

$$\begin{aligned} Z_1 &= \mathrm{Softmax}_{\mathrm{rowwise}}\left(\frac{O_1 O_2^T}{\sqrt{d}}\right) O_1 \\ Z_2 &= \mathrm{Softmax}_{\mathrm{columnwise}}\left(\frac{O_2 O_1^T}{\sqrt{d}}\right) O_2. \end{aligned} \tag{5}$$

As shown in (5), rowwise and columnwise normalizations are achieved by transposing the similarity matrix, and meanwhile, the computational complexity is also reduced because the similarity matrix only needs to be calculated once. In this way,

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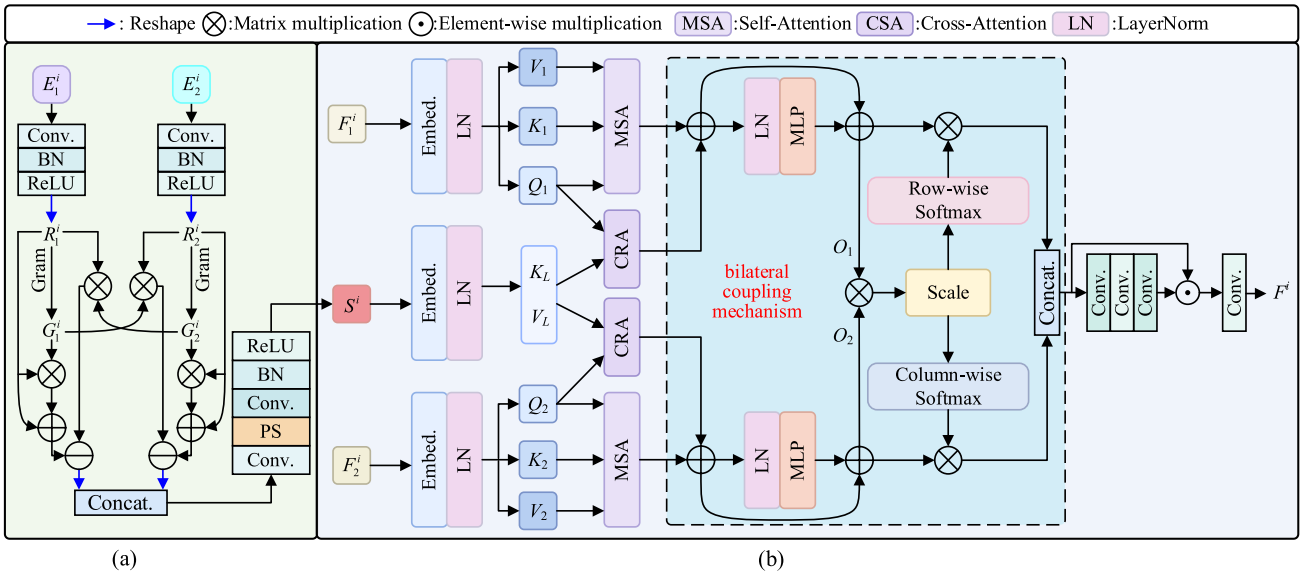


Fig. 3. Schematic of our designed DALM (a) SpeAB and (b) SpaAB.

the bitemporal features are coupled better due to the simultaneous refinement along different dimensions. Subsequently, the refined features Z_1 and Z_2 are concatenated and mapped by a convolution block to obtain the integrated feature F^i at the ith scale. The multiscale features are then fed into the MSFD to predict the change map.

D. Optimization

For the training of the CD network, a weighted cross-entropy loss is considered

$$\mathcal{L}_{\mathrm{CD}} = -\left(\alpha M \odot \log \hat{M} + \beta (1-M) \odot \log\left(1-\hat{M}\right)\right) \tag{6}$$

where $M \in \mathbb{R}^{rh \times rw}$ and $\hat{M} \in \mathbb{R}^{rh \times rw}$ are the ground truth and the predicted change map, respectively. Considering the imbalance between foreground and background regions in the CD task, α and β are adjusted to ensure that rare changed pixels are adequately emphasized during optimization. Here, we empirically set α and β to 0.8 and 0.2, respectively.

Finally, the total loss is expressed as the sum of the SR loss $\mathcal{L}_{\mathrm{SR}}$ and the CD loss $\mathcal{L}_{\mathrm{CD}}$ for the joint learning of the proposed SSDA-Net

$$\mathcal{L}_{\mathrm{Total}} = \mathcal{L}_{\mathrm{SR}} + \mathcal{L}_{\mathrm{CD}}. \tag{7}$$

IV. Experiments and Result Analysis

In this section, we first introduce the three datasets used in this article, followed by detailed descriptions in terms of implementation details. Then, experimental comparisons and analysis of all methods are conducted on the three considered datasets. In addition, extensive ablation experiments are also performed to verify the effectiveness of the proposed SSDA-Net.

A. Datasets

1) SD-CSCD: To facilitate CSCD in remote sensing imagery, we construct a new dataset, SD-CSCD, which reflects substantial changes within Shandong Province, China, in terms of urban expansion, infrastructure development, and land-use transformation. In this dataset, bitemporal remote sensing images were acquired by two satellites: the 2-m-resolution GF-6 and the 1-m-resolution GF-2, which exhibit notable differences in terms of spectral response and spatial resolutions. The resolution ratio between them is 2. The bitemporal images were acquired in 2019 and 2024.

To address the significant challenges inherent in cross-satellite change detection, particularly in mitigating the domain shifts caused by heterogeneous sensors and varying imaging conditions, we constructed the SD-CSCD dataset. Grounded geographically in Shandong Province, this dataset provides a robust benchmark for developing algorithms capable of aligning and detecting changes across diverse satellite sources.

As detailed in Fig. 4, SD-CSCD encompasses six distinct semantic categories tailored to complex land cover transformations: general change, industrial facilities, scattered buildings, facility agriculture, road networks, and water bodies. The dataset exhibits a distribution designed to reflect real-world scenarios. It is heavily populated by 1800 general change image pairs and 1500 pairs containing industrial facilities, which highlight the dynamic development of the focused regions. In addition, 1300 scattered building image pairs and 1200 facility agriculture image pairs offer rich textural diversity for the dataset. Finally, the dataset includes 1000 samples of road networks and 480 samples of water bodies. Collectively, these categories form a comprehensive and heterogeneous data foundation, specifically constructed to advance research in CSCD.

All images are first preprocessed by ENVI, applying radiometric calibration, atmospheric correction, orthorectification, and geometric refinement based on RPC models. Second, the multispectral images from the two satellites are pansharpened by the NNDiffuse tool in ENVI for resolution improvement. Then, the pansharpened multispectral images are coregistered. Next, the registered images are cropped into patches for convenient labeling and subsequent training. It should be noted that, in order to better match existing networks, this dataset

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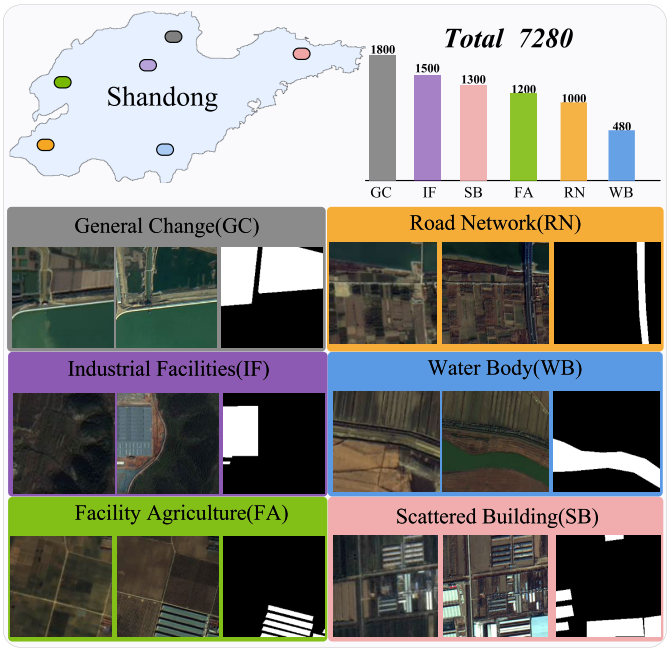


Fig. 4. Overview of the proposed SD-CSCD dataset.

uses only the R, G, and B bands to form image pairs, even though the multispectral images acquired by the GF-6 and GF-2 satellites also contain the near-infrared band. Considering the resolution difference, the preevent images from GF-6 satellite are divided into patches with a size of 128 × 128, while the postevent images from GF-2 satellite are partitioned into 256 × 256 image blocks. Finally, **7 280** pairs of image patches are obtained and are split into training, validation, and testing datasets with a ratio of 7:1:2, ensuring effective model performance evaluation.

2) DE-CD: The DE-CD dataset is derived from the existing DynamicEarthNet dataset, including PlanetScope images at 3 m/pixel and Sentinel-2 images at 10 m/pixel [22]. The Sentinel-2 images from 2018 are used as the preevent data, while PlanetScope images captured one year later serve as the postevent data. Thus, the spatial-resolution ratio between Sentinel-2 and PlanetScope imagery is 3.3. In addition, the three bands, R, G, and B, are considered in this dataset. After removing samples without notable changes, 506 image pairs are obtained and randomly divided into 354 for training, 51 for validation, and 101 for testing. For more efficient model training, each image in this dataset is further cropped into nonoverlapping 256 × 256 patches.

3) LEVIR-CD: The LEVIR-CD dataset is a large-scale public benchmark designed for building CD. It comprises 637 remote sensing image pairs from Google Earth with a spatial resolution of 0.5 m/pixel and a size of 1024 × 1024. Similarly, we also segment these images into patches with a size of 256 × 256. Therefore, 10 192 image pairs are generated, of which 7120 pairs are used for training, 1024 pairs for validation, and 2048 pairs for testing. To construct the cross-resolution dataset, the preevent image in the LEVIR-CD dataset is further downsampled by a factor of 2 and regarded as the LR image.

B. Implementation Details

In the following experiments, the proposed SSDA-Net is compared with nine methods, including FC-EF [55], FC-Siam-conc [55], FC-Siam-diff [55], ChangeFormer [56], BIT [57], HANet [58], SNUNet [24], SILI-CD [21], and DCIL-Net [23]. To quantitatively evaluate the results of all methods, we employ five commonly-used metrics: precision (P), recall (R), $F1$-score ($F1$), mean intersection over union (mIoU), and overall accuracy (OA) [59]. For these metrics, higher values mean better CD performance. The main characteristics of these methods are briefly summarized below.

FC-EF: It is a deep learning-based CD method that concatenates multitemporal RS images as the network input and utilizes a fully convolutional network for CD.

FC-Siam-conc: This method adopts a Siamese network architecture and performs the inference of changed regions by concatenating bitemporal features.

FC-Siam-diff: Multitemporal RS image features are independently extracted by a Siamese fully convolutional network with shared weights, and the absolute difference between the features is used to predict the CD result.

ChangeFormer: It is a transformer-based Siamese network architecture that combines a hierarchical transformer encoder with an MLP decoder to effectively capture multiscale and long-range information.

BIT: This network effectively combines CNNs and transformers. In this method, semantic tokens are generated through segmentation, and cross-attention is introduced to improve CD performance.

HANet: This network integrates a lightweight self-attention mechanism with a dense attention block to fuse multiscale features, thereby further refining feature representations.

SNUNet: This network is a densely connected Siamese architecture that strengthens intermediate feature representations and exploits a channel attention module to model contextual information.

SILI-CD: The method is a cross-resolution CD network that synthesizes bitemporal samples with random resolution differences, aligns multilevel Siamese features via local-window self-attention, and employs an INR-based MLP decoder to infer HR change maps.

DCIL-Net: It is a dual-branch cross-resolution CD network that combines image-space SR alignment with feature-space interactive correction to jointly exploit HR and LR bitemporal information for more accurate change mapping.

The proposed SSDA-Net is trained on a single NVIDIA RTX 3090 GPU using the SGD optimizer. We set the tradeoff parameter λ in (1) as 0.1. The batch size is set to 4. Following the widely adopted settings in most CD methods, the number of training epochs is set to 200. Meanwhile, the initial learning rate is set to 0.001, and the learning rate drops to one-tenth of its original value every 100 epochs. For all compared methods, the batch size is 4.

C. Comparison Experiment With State-of-the-Art Methods

1) Results on the SD-CSCD Dataset: Table I summarizes the performance comparison of different methods in terms of

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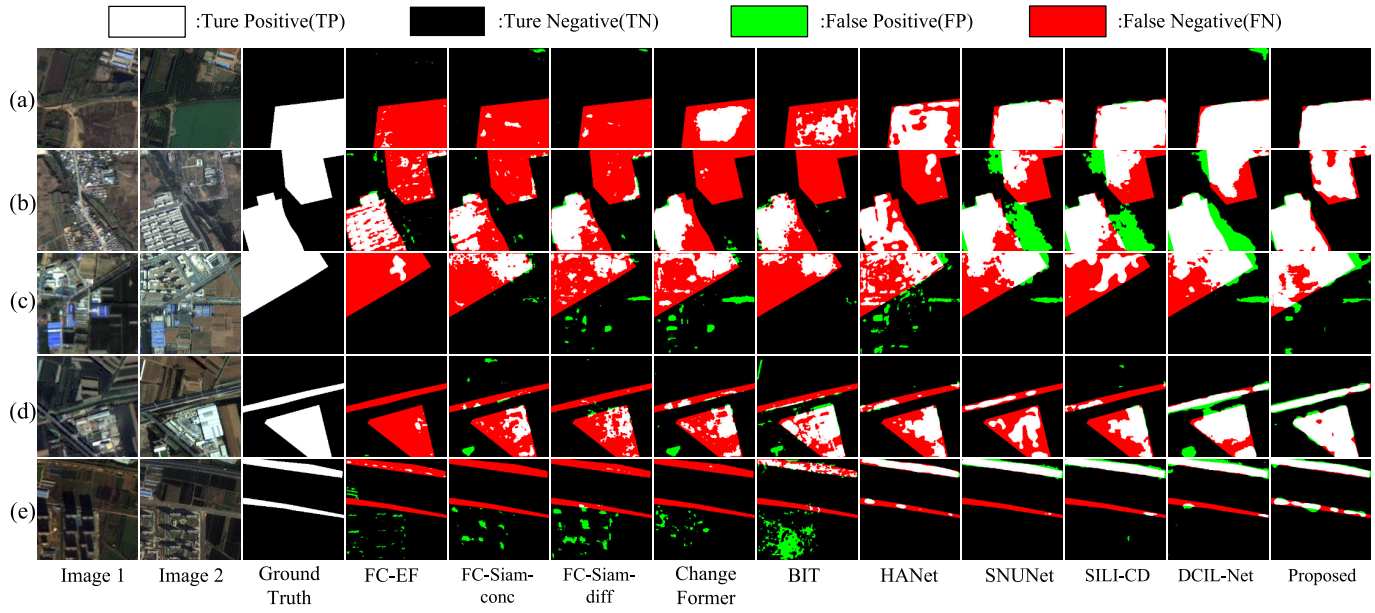


Fig. 5. Qualitative comparison of the CD maps produced by all methods on some examples of images from the SD-CSCD dataset. (a)–(e) Prediction results of all methods on different image pairs.

TABLE I

QUANTITATIVE RESULTS (%) OBTAINED BY ALL METHODS ON THE SD-CSCD DATASET (BEST VALUES ARE IN BOLD)

Method	P	R	F1	mIoU	OA
FC-EF	38.77	30.91	34.40	58.43	96.57
FC-Siam-conc	48.32	42.07	44.98	62.99	96.79
FC-Siam-diff	52.27	35.58	42.34	70.44	97.17
ChangeFormer	59.55	36.65	45.37	63.37	97.43
BIT	53.57	44.18	48.42	64.58	97.25
HANet	61.46	53.77	57.36	68.09	97.26
SNUNet	62.37	62.61	62.49	69.16	96.32
SILI-CD	64.42	63.38	63.90	71.83	97.88
DCIL-Net	63.69	64.71	64.20	71.73	97.86
Proposed SSDA-Net	**76.31**	**77.23**	**76.77**	**74.03**	**98.23**

TABLE II

QUANTITATIVE RESULTS (%) OBTAINED BY ALL METHODS ON THE DE-CD DATASET (BEST VALUES ARE IN BOLD)

Method	P	R	F1	mIoU	OA
ChangeFormer	28.16	32.37	30.12	58.36	99.06
BIT	31.06	32.46	31.74	59.02	99.17
HANet	35.45	32.96	34.16	60.03	99.29
SNUNet	37.73	35.73	36.70	57.47	99.21
SILI-CD	50.96	42.76	46.50	61.07	99.39
DCIL-Net	59.81	42.47	49.67	**65.26**	**99.48**
Proposed SSDA-Net	**60.33**	**43.05**	**50.25**	64.43	99.44

all considered evaluation metrics on the SD-CSCD dataset. The best values are marked in bold. We can see that the proposed SSDA-Net achieves the best performance across all five metrics. Notably, compared with the second-best method, SILI-CD, our SSDA-Net yields improvements of 1.30% in mIoU. These results demonstrate that the proposed method can better alleviate the influences of inherent spectral and spatial discrepancies in cross-satellite multitemporal images, thereby exhibiting accurate CD capability.

Fig. 5 displays the visual results of all methods on some examples from the SD-CSCD dataset, where white, black, green, and red pixels denote true positive (TP), true negative (TN), false positive (FP), and false negative (FN), respectively. Since the multitemporal images in the SD-CSCD dataset are captured by different imaging sensors having distinct spectral responses and spatial characteristics, FC-EF, FC-Siam-conc, and FC-Siam-diff produce poor results in regions with extensive changes, as shown in Fig. 5(a)–(c). Meanwhile, ChangeFormer, BIT, and HANet can only detect a small fraction of these change areas. For the cross-resolution methods, such as SILI-CD and DCIL-Net, although their results are better than those of ChangeFormer and BIT, they incur a nonnegligible number of FP and FN. In addition, Fig. 5(e) shows that the proposed SSDA-Net can more efficiently suppress pseudochanges caused by shadow of buildings, while other methods fail to locate the newly built road regions. Overall, on the SD-CSCD dataset, the proposed SSDA-Net obtains the best detection performance due to the SpeAB and SpaAB in DALMs.

2) Results on the DE-CD Dataset: In this section, we compare the performance of all methods on the DE-CD dataset. Considering the limited fitting ability of FC-EF, FC-Siam-conc, and FC-Siam-diff, we adopt the same strategy as [21] and do not consider these three methods on this dataset. Table II reports the evaluation metrics of remaining methods. All methods show limited performance on this dataset. This may be because this dataset is derived from datasets for other tasks. Compared to other methods, our proposed SSDA-Net attains the best results in terms of P, R, and $F1$. Although DCIL-Net provides the best mIoU and OA values, our method produces the second-best results in both metrics. Considering the overall performance, the proposed SSDA-Net demonstrates more accurate detection capability for both changed and unchanged regions.

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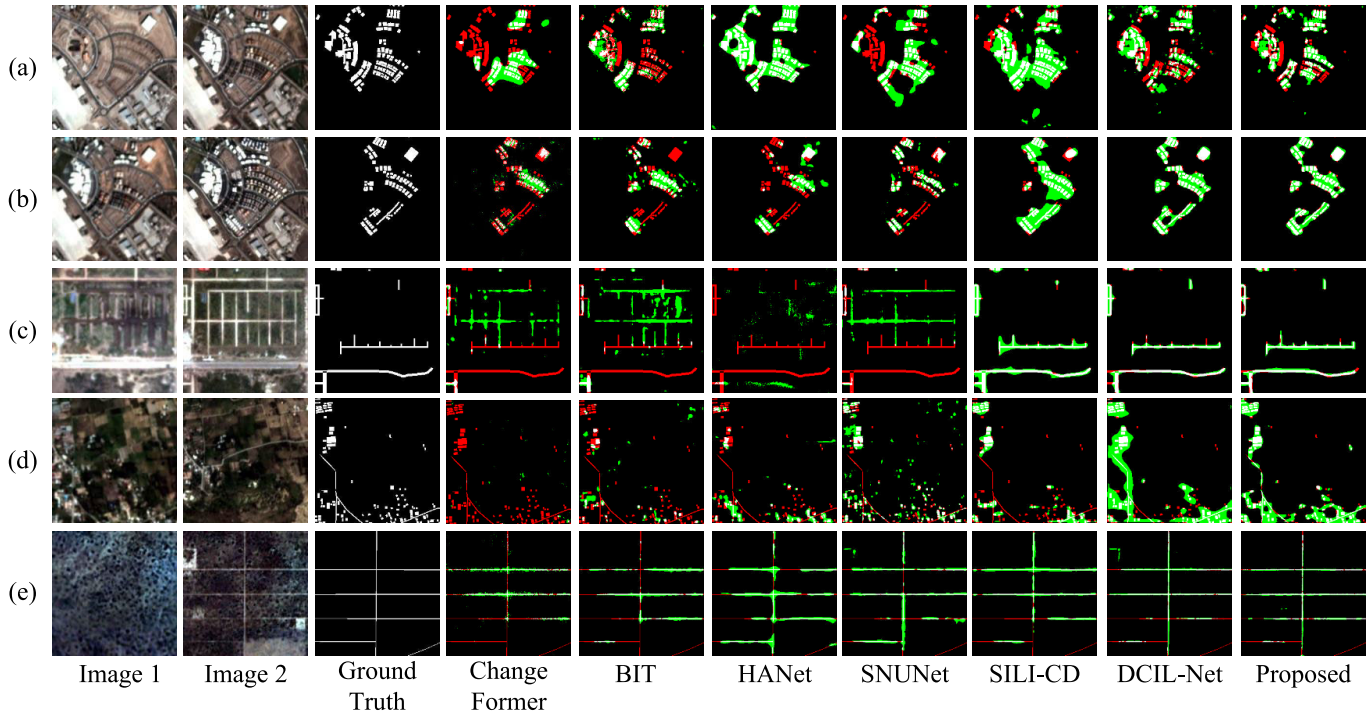


Fig. 6. Qualitative comparison of the CD maps produced by all methods on some examples of images from the DE-CD dataset. (a)–(e) Prediction results of all methods on different image pairs.

TABLE III
QUANTITATIVE RESULTS (%) OBTAINED BY ALL METHODS ON THE LEVIR-CD DATASET (BEST VALUES ARE IN BOLD)

Method	P	R	F1	mIoU	OA
FC-EF	70.55	73.65	72.07	66.76	96.84
FC-Siam-conc	57.71	74.21	64.93	70.46	96.17
FC-Siam-diff	60.75	74.83	67.06	68.91	93.77
ChangeFormer	62.35	90.03	73.68	76.51	95.06
BIT	61.16	93.55	73.96	77.28	96.64
HANet	86.61	87.36	86.98	79.95	98.23
SNUNet	86.71	88.21	87.45	77.71	98.06
SILI-CD	90.27	86.58	88.39	82.87	97.32
DCIL-Net	91.67	88.75	90.19	85.03	99.42
Proposed SSDA-Net	**92.55**	**90.36**	**91.44**	**86.43**	**99.49**

From Fig. 6(a) and (b), we can see that the proposed SSDA-Net performs better in the change areas of dense buildings, while more FP regions can be found in the results of SILI-CD and DCIL-Net. Besides, as shown in Fig. 6(c), ChangeFormer, BIT, HANet, and SNUNet exhibit more obvious FN errors along road networks, failing to capture many subtle structural changes. The reason for this may be that transformer-based methods cannot efficiently model the change information in small local areas. Compared with these methods, the proposed SSDA-Net generates more accurate and visually consistent change maps, which are also more accurate than those produced by existing cross-resolution CD methods. This is attributed to the dual-resolution joint learning mechanism and the spectral–spatial alignment strategy.

3) Results on the LEVIR-CD Dataset: Table III reports the quantitative results of the proposed SSDA-Net compared with nine state-of-the-art methods on the LEVIR-CD dataset. As shown in the table, our SSDA-Net achieves the highest performance across all five evaluation metrics with scores of 92.55%, 90.36%, 91.44%, 86.43%, and 99.49%, respectively, for P, R, $F1$, mIoU, and OA. Overall, this confirms the effectiveness of our designed LR and HR joint learning framework.

To provide a more intuitive comparison, Fig. 7 presents visual examples of CD results on five image pairs. As shown in Fig. 7(a)–(c), BIT suffers from limited performance in terms of suppressing the unchanged areas, while the results of DCIL-Net contain larger FN regions. Compared to other methods, we can see that the CD maps predicted by SSDA-Net exhibit closer contours to the ground truth. The image pair in Fig. 7(e) contains small changed areas. It can be seen that HANet and SNUNet cannot efficiently preserve the changed areas of small objects. Our method generates more refined results and outlines the changed objects better. Therefore, the qualitative analysis in Fig. 7 further confirms that SSDA-Net can accurately capture true changed regions while effectively suppressing FP and FN pixels.

D. Ablation Study

In this section, to validate the effectiveness of the individual modules within the proposed SSDA-Net, we conduct comprehensive ablation experiments across the SD-CSCD and LEVIR-CD datasets. These experiments aim to investigate the influences of SpeAB, SpaAB, and backbone on the CD performance of different satellites. Table IV evaluates the contributions of all these modules and provides the corresponding values of the evaluation metrics. In this table, w/o indicates that the corresponding module is removed from the complete SSDA-Net. Moreover, Figs. 8 and 9 present examples of CD maps of different versions of SSDA-Net on the two datasets.

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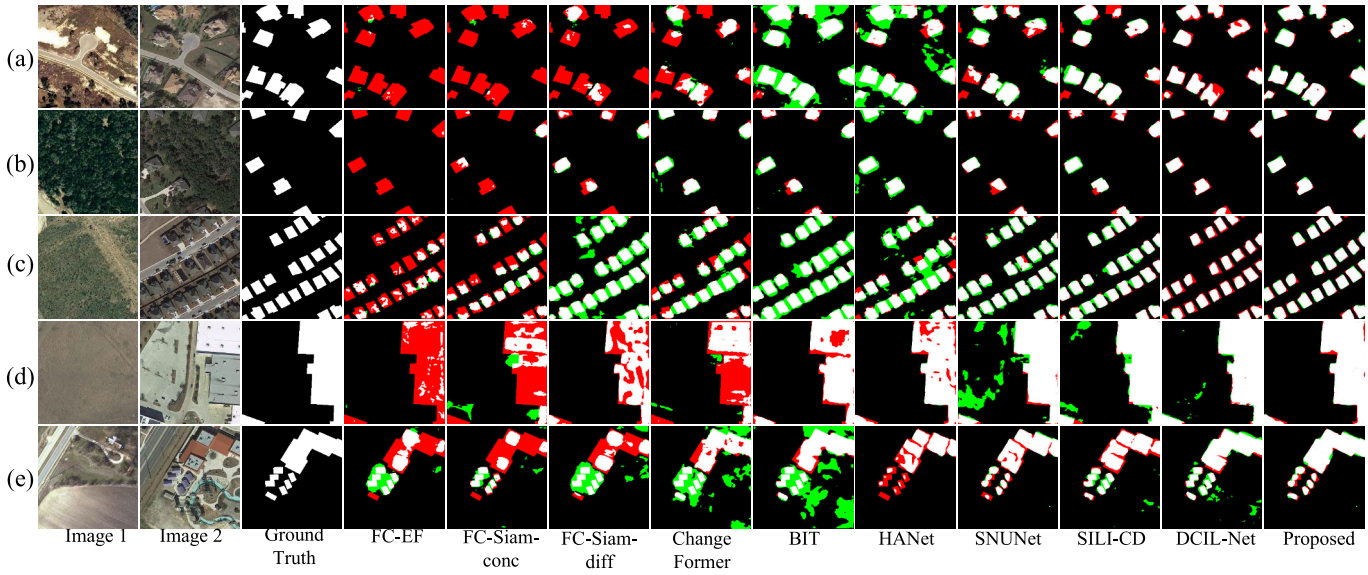


Fig. 7. Qualitative comparison of all methods on the LEVIR-CD dataset. (a)–(e) Prediction results of all methods on different image pairs.

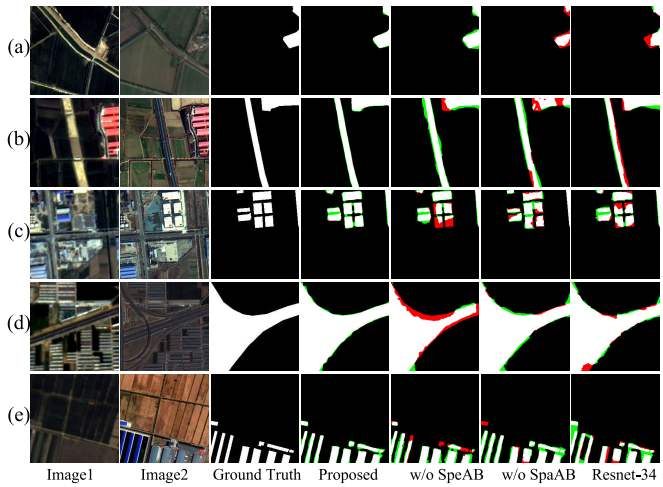


Fig. 8. Qualitative results of the ablation study on the SD-CSCD dataset. (a)–(e) Prediction results of all methods on different image pairs.

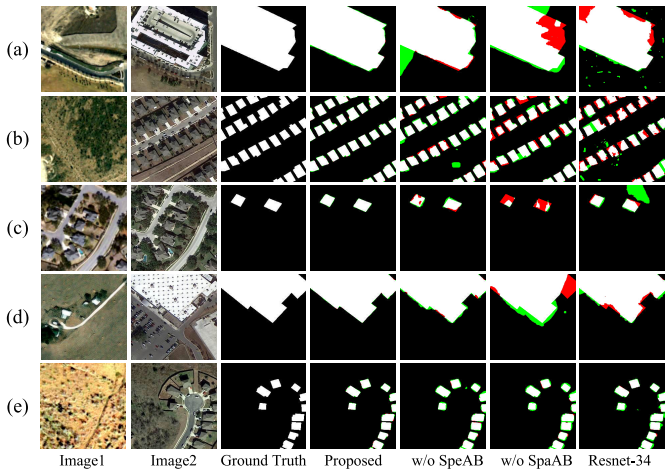


Fig. 9. Qualitative results of the ablation study on the LEVIR-CD dataset. (a)–(e) Prediction results of all methods on different image pairs.

1) Ablation of SpeAB: As shown in Table IV, removing the SpeAB module leads to obvious degradation across all metrics when compared with the complete SSDA-Net. Specifically, for the SD-CSCD dataset, P and R decrease by 3.36% and 4.97%, respectively. Furthermore, the other three metrics also significantly drop. These results indicate that the SpeAB module improves the balance between FPs and FNs, thereby yielding higher $F1$ and mIoU. By analyzing Figs. 8 and 9, we can also see that a large number of pixels are misclassified and the structures of buildings cannot be captured effectively when SpeAB is removed. So, the spectral alignment on LR features is crucial for improving the overall robustness on the CSCD task.

2) Ablation of SpaAB: Table IV shows that negative influences arise on all metrics when we ablate the SpaAB module. For example, the P and R values on the LEVIR-CD dataset decrease by 1.12% and 1.41%, respectively. Both the $F1$ score and mIoU are also reduced. This indicates that the absence of the SpaAB module prevents SSDA-Net from fully leveraging the spatial features in cross-satellite multitemporal images. Meanwhile, similar performance can be observed in Figs. 8 and 9. For example, SSDA-Net performs effectively when dealing with small and densely-distributed change areas in Figs. 8(c) and 9(b). However, when the SpaAB module is

TABLE IV

ABLATION STUDY OF SPEAB, SPAAB, AND BACKBONE ON THE SD-CSCD AND THE LEVIR-CD DATASETS

Datasets	Settings	P	R	F1	mIoU	OA
SD-CSCD	w/o SpeAB	72.95	72.26	72.59	72.65	98.01
	w/o SpaAB	72.37	71.85	72.11	73.88	97.81
	ResNet-34	74.18	75.37	74.77	**74.75**	97.67
	Proposed	**76.31**	**77.23**	**76.77**	74.03	**98.23**
LEVIR-CD	w/o SpeAB	90.93	88.79	89.85	82.21	96.37
	w/o SpaAB	91.43	88.95	90.17	83.62	97.21
	ResNet-34	91.82	90.03	91.40	85.83	98.61
	Proposed	**92.55**	**90.36**	**91.44**	**86.43**	**99.49**

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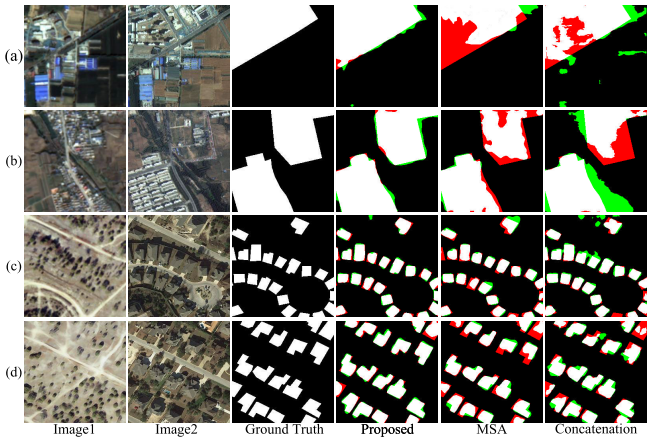


Fig. 10. Qualitative results of different spatial attention mechanisms in SpaAB. (a) and (b) Prediction results on the SD-CSCD dataset. (c) and (d) Prediction results on the LEVIR-CD dataset.

TABLE V

COMPARISON OF THE ATTENTION AND THE BILATERAL COUPLING MECHANISM ON THE SD-CSCD AND THE LEVIR-CD DATASETS

Datasets	Settings	P	R	F1	mIoU	OA
SD-CSCD	MSA	74.72	75.91	75.31	72.38	95.22
	Concatenation	74.37	76.98	75.65	73.67	98.14
	Ours	**76.31**	**77.23**	**76.77**	**74.03**	**98.23**
LEVIR-CD	MSA	91.52	88.55	90.01	84.61	97.86
	Concatenation	91.98	89.83	91.24	85.07	98.95
	Ours	**92.55**	**90.36**	**91.44**	**86.43**	**99.49**

removed, the obtained results suffer from inferior performance in detecting the boundaries of change areas.

3) Ablation of Backbone: To further investigate the impact of different backbone networks on CD performance, the ResNet-18 encoder is replaced with ResNet-34. When employing ResNet-34 as the encoder, the mIoU value on the SD-CSCD dataset increases compared with ResNet-18, whereas all other evaluation metrics show a decline. ResNet-18 contains 11.69M parameters and requires 1.81 GFLOPs, while ResNet-34 nearly doubles both the parameter count and computational complexity with 21.80M parameters and 3.66 GFLOPs. This may be attributed to the larger parameter scale of ResNet-34, which hinders sufficient training under the limited data conditions. On the LEVIR-CD dataset, the introduction of ResNet-34 results in a decrease of all metrics. This is confirmed by the qualitative results in Figs. 8 and 9, which reveal an increase of FP and FN regions when the backbone network is replaced.

E. Investigation of SpaAB

In this part, we present more investigations on the SD-CSCD and LEVIR-CD datasets to analyze the attention mechanism in SpaAB. Table V and Fig. 10 present the quantitative and qualitative results, respectively. First, we investigate the influence of CRA. Specifically, we replace CRA with a standard MSA operating solely on HR features while keeping the remaining components unchanged. In Table V, we can observe that the MSA strategy yields poor numerical performance on SD-CSCD and LEVIR-CD datasets, indicating that the vanilla attention mechanism cannot fully exploit the complementary information in LR and HR features. In Fig. 10, the predicted change maps on these datasets exhibit fuzzy and irregular object contours and an obvious increase of FP and FN areas along building roofs when MSA is adopted.

Moreover, we also validate the effectiveness of the bilateral coupling mechanism in SpaAB. In this experiment, we simply concatenate the outputs of the two HR branches, instead of computing the similarity matrix between the enhanced bitemporal HR features. Then, the concatenated feature is mapped by a convolution block to generate the fused representation. The results in Table V demonstrate that the introduction of the bilateral coupling mechanism brings improvements in terms of P, R, and $F1$. For visual analysis, we can see that the concatenation operation leads to worse detection results because the bitemporal features do not interact sufficiently. Therefore, all these results denote that the designed bilateral coupling mechanism is more effective than either vanilla attention or simple feature concatenation.

TABLE VI

IMPACT OF DIFFERENT SR MODULES ON THE SD-CSCD DATASET WITH THE SSDA-NET CD NETWORK

Method	PSNR	NIQE	P	R	F1	mIoU	OA
DCIL-Net	26.07	3.88	63.69	64.71	64.20	71.73	97.86
Bilinear+our CD	25.78	4.02	65.98	65.21	65.59	70.65	96.44
SRM+our CD	26.54	3.61	76.02	75.95	75.98	73.88	97.92
SSDA-Net	**27.93**	**3.56**	**76.31**	**77.23**	**76.77**	**74.03**	**98.23**

F. Effect of the SR Network

In this section, we study the impact of different SR strategies in SSDA-Net on the SD-CSCD dataset. Specifically, we compare two variants: 1) an alternative SR network, such as the SR module (SRM) used in DCIL-Net and 2) bilinear interpolation. In the latter case, the LR feature extraction is implemented on the interpolated images by a multiscale CNN. For a comprehensive evaluation, the SR performance of different strategies is also provided. Here, we utilize the no-reference image quality assessment metric, NIQE [60], to measure the quality of the super-resolved images from Time 1 owing to the absence of reference images. On the contrary, PSNR is used as the full-reference image quality assessment metric to evaluate the super-resolved images from Time 2.

Table VI lists all results of SR and CD metrics on the SD-CSCD dataset. Meanwhile, in Table VI, we also add the results of DCIL-Net for direct comparison. As summarized in Table VI, the bilinear interpolation operation yields the worst performance, confirming that simple upsampling is insufficient to recover necessary details for CD. Moreover, compared to DCIL-Net, numerical improvements in terms of P, R, mIoU, and OA are observed when we integrate the SRM into our proposed method. Thus, the improved results verify the effectiveness of our constructed dual-resolution joint learning framework. Furthermore, although different SR strategies have some influence on SSDA-Net, we can also observe that our designed framework still has good performance when

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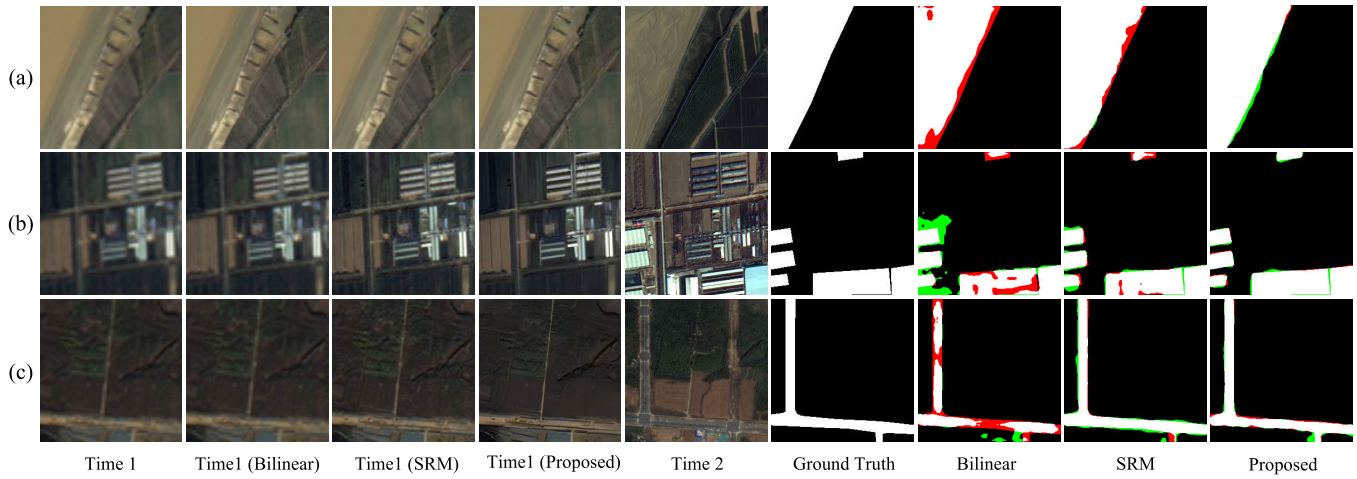


Fig. 11. Qualitative comparisonof different SR strategies for SSDA-Net on the SD-CSCD dataset. (a)–(c) Different bitemporal image pairs.

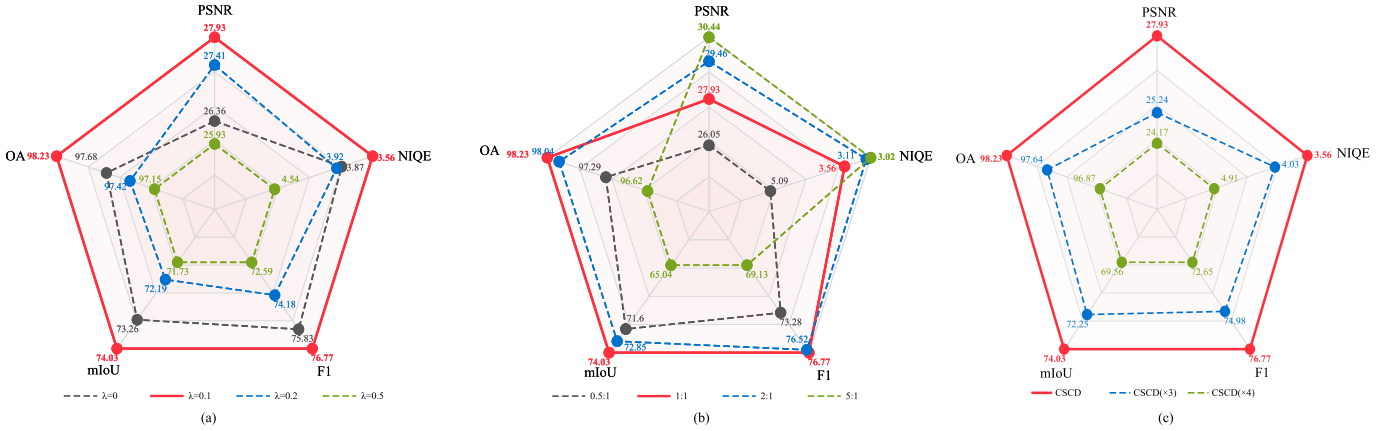


Fig. 12. Analysis of key parameters and resolution ratios on the SD-CSCD dataset. (a) Influence of λ in the SR loss. (b) Influence of the relative weights between the SR and CD losses. (c) Influence of different resolution ratios. For NIQE, lower values indicate better image quality, and inverse normalization is used for visualization.

compared with the results in Table I. This further demonstrates the robustness of our method against spectral and spatial differences among different satellites.

In addition to the CD results, Fig. 11 also shows the SR results of different strategies. In the SR images provided by our method, more details and clear spatial structures can be perceived, while other SR images suffer from slight blurring effects. For CD performance, more FN and FP pixels are generated by the variants of SSDA-Net because of the lack of detailed information. More refined prediction maps are provided by the complete SSDA-Net.

G. Parameter Analysis

To further analyze the influence of key training parameters and the applicability of SSDA-Net under different resolution gaps, we conduct additional experiments on the SD-CSCD dataset. The results are shown in Fig. 12.

We investigate the effect of λ in the SR loss. As shown in Fig. 12(a). The parameter λ controls the contribution of the degradation-consistency constraint for the Time-1 LR image. When $\lambda = 0$, this constraint is removed, and the model obtains PSNR, NIQE, $F1$, mIoU, and OA values of 26.36, 3.87, 75.83%, 73.26%, and 97.68%, respectively. When $\lambda = 0.1$, all metrics achieve the best performance, with PSNR, NIQE, $F1$, mIoU, and OA values of 27.93, 3.56, 76.77%, 74.03%, and 98.23%, respectively. This indicates that a proper degradation-consistency constraint helps the SR network adapt to the Time-1 cross-satellite image distribution. However, when λ is further increased to 0.2 or 0.5, both SR quality and CD accuracy decrease, suggesting that an excessively large degradation constraint may weaken the supervised reconstruction learning and degrade the downstream CD performance. Therefore, $\lambda = 0.1$ is adopted in our experiments.

Fig. 12(b) presents the analysis of the relative weights of the SR and CD losses. Specifically, we fix the weight of the CD loss to 1 and evaluate different weights of the SR loss, including 0.5, 1, 2, and 5. The setting $\mathcal{L}_{SR} : \mathcal{L}_{CD} = 1:1$ achieves the best CD performance. When the SR weight is reduced to 0.5, insufficient SR supervision leads to degraded SR quality and CD accuracy. When the SR weight is increased to 2 or 5, PSNR and NIQE are improved, but $F1$, mIoU, and OA decrease. In particular, when $\mathcal{L}_{SR} : \mathcal{L}_{CD} = 5:1$, $F1$ and mIoU drop to 69.13% and 65.04%, respectively. This

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TABLE VII
COMPARISON OF JOINT TRAINING AND SEPARATE TRAINING ON THE SD-CSCD DATASET

Method	P	R	F1	mIoU	OA
Separate	75.14	74.73	74.93	72.81	98.17
Joint (Ours)	**76.31**	**77.23**	**76.77**	**74.03**	**98.23**

demonstrates that overemphasizing the SR objective makes the model focus more on image reconstruction while weakening its ability to learn change-discriminative features. Thus, the equal-weight setting provides the best tradeoff between SR quality and CD accuracy.

H. Effect of Resolution Ratio

We evaluate the influence of different resolution ratios. In addition to the original SD-CSCD setting with a resolution ratio of 2, we construct two additional settings with resolution ratios of 3 and 4, denoted as SD-CSCD ($\times 3$) and SD-CSCD ($\times 4$), respectively. As the resolution ratio increases from 2 to 3 and 4, PSNR decreases from 27.93 to 25.24 and 24.17, while NIQE increases from 3.56 to 4.03 and 4.91. As shown in Fig. 12(c), the CD performance shows the same trend: $F1$ decreases from 76.77% to 74.98% and 72.65%, mIoU decreases from 74.03% to 72.25% and 69.56%, and OA decreases from 98.23% to 97.64% and 96.87%. These results indicate that larger resolution gaps introduce more severe spatial information loss and make cross-resolution feature alignment more difficult. Nevertheless, SSDA-Net still maintains reasonable performance under the $\times 3$ and $\times 4$ settings, showing its applicability to moderate-to-large resolution differences.

I. Effect of Joint Training Strategy

To investigate the necessity of the proposed joint training strategy, we compare it against a degraded variant termed Separate, in which the SR network is first trained independently and then frozen as a fixed inference module during CD training. As reported in Table VII, the joint training strategy consistently outperforms the Separate variant across all metrics. Specifically, the Separate variant yields decreases of 1.84% in $F1$ and 1.22% in mIoU. Notably, R drops by 2.50% while P decreases by only 1.17%, indicating that freezing the SR network weakens the LR feature guidance in DALMs and causes more missed detections. These results confirm that joint training enables SR features to be implicitly optimized toward change-relevant representations, facilitating synergistic optimization of the SR and CD networks.

V. CONCLUSION

In this article, we have proposed SSDA-Net to address the mismatches in terms of spectral response and spatial resolutions caused by heterogeneous imaging sensors in CD from different satellites. The proposed SSDA-Net is composed of an SR network and a CD network. Within the CD network, we develop DALMs for joint learning of LR and HR features. Specifically, the SpeAB in DALM leverages Gram-matrix-based style modeling to align the spectral distributions in bitemporal LR features to improve spectral consistency across satellites. In addition, to mitigate the misalignment induced by spatial resolution differences, we introduce the SpaAB, which imposes a bilateral coupling mechanism on HR features for better fusion of bitemporal features. Furthermore, we construct a VHR cross-satellite CD dataset, SD-CSCD, providing a reliable benchmark for evaluating CD methods in realistic cross-satellite scenarios. Experiments on three CD datasets with different characteristics demonstrate that the proposed SSDA-Net delivers superior performance compared with nine state-of-the-art methods. For future work, we will explore more efficient dual-resolution modeling schemes to deal with more complex spectral and spatial property differences among diverse imaging sensors.

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